SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Dr.V.PARTHIPAN

Code of Conduct:

I C. PRIYANKA / 192511074 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. PRIYANKA

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → ________ → Transport → Network → ________ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

ANSWERS

Question 1 Missing Layers: Session and Data Link.
Concept Map:
Application → Presentation → Session → Transport → Network → Data Link → Physical Explanation:

- **Session Layer establishes, manages and terminates communication sessions.**
- **Data Link Layer provides framing, MAC addressing and error detection.**
- **These layers ensure reliable communication by maintaining sessions and detecting transmission errors.**

Question 2
Concept Map:
Transport → Segment Network → Packet Data Link → Frame Physical → Bits
Analysis:
During encapsulation, application data becomes Segments, then Packets, Frames and finally Bits for transmission. At the receiver, decapsulation converts Bits back into Frames, Packets, Segments and application data.

Question 3
Concept Map:
Data Link → MAC Address → Local Delivery Network → IP Address → Global Delivery
Analysis: The MAC address delivers data within the local network, while the IP address routes data across different networks. Together they ensure complete end-to-end delivery.

Question 4
Concept Map:
Data Link Layer → Error Detection Transport Layer → Error Correction
Analysis:
The Data Link Layer detects errors using CRC/checksum and retransmits damaged frames on a local link. The Transport Layer ensures reliable end-to-end delivery using acknowledgements, sequencing and retransmission, improving communication reliability.

Question 5
Concept Map:
Application (Browser) → Presentation (Encryption/SSL) → Session → Transport (TCP) → Network (IP) → Data Link (Ethernet/Wi-Fi) → Physical (Cable/Wireless)
Analysis:
Failure of any layer interrupts communication. For example, a Physical Layer failure disconnects the network, while a Transport Layer failure prevents reliable webpage loading.

Question 6
Analysis:
Physical and Data Link layers are working correctly, but the Network Layer has failed. Because the Network Layer cannot route packets, the Transport and Application layers also fail.

Troubleshooting: • Check IP address configuration.

- **Verify router and gateway settings.**
- **Test routing using ping/traceroute.**
- **Restore Network Layer connectivity to recover upper-layer communication.**

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)

Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation